

# Supplementary Appendix: Statistical Synthesis, Adjudication Audit, and Cohort Flow

**Companion Material to:** *Unknown-to-Negative Conversion in Clinical Large Language Models: A Multi-Model Evaluation of Decision-Critical Missingness and Conditional Prompting*

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**Repository:** <https://github.com/dochobbs/aom-chart>

## Section S1: Experimental Cohort Architecture & Study Flow

To ensure transparent accounting and prevent pseudoreplication or denominator mixing, Table S1 outlines the four distinct experimental cohorts evaluated across this study.

**Table S1: Consolidated Study Cohort Accounting (Unique Traces vs. Turns)**

| Cohort ID                                     | Description / Arm                                                                                                                                                                                                                                         | Clinical Cases  | Models Evaluated                                        | Unique Traces (Total Turns)          | Primary Analytical Endpoints                                                                                                                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cohort 1: Foundational AOM Battery</b>     | Unique generations: Baseline catalog ( $N = 140$ ) + pre-specified identity contrasts ( $N = 228$ ) + single-sentence prompt intervention ( $N = 56$ ). ( <i>Turn 2 sealed probing was evaluated on 192 multi-turn responses from the identity runs</i> ) | aom (24mo)      | 10 models (4 vendor families)                           | <b>424 unique traces</b> (652 turns) | Closed-world fabrication; verbosity association; model signatures; learned association errors; dual-pass scoring vs. clinician adjudication; 0/228 prescription invariance; 187/192 Turn 2 recognition |
| <b>Cohort 2: Factorial Occupation Battery</b> | $2 \times 2$ factorial (Mother/Father $\times$ Nurse/Unemployed) at $N = 6$ replicates per cell across 2 cases                                                                                                                                            | aom , head_24mo | 6 models ( fable , sonnet , opus , flash , pro , grok ) | <b>288 unique traces</b> (576 turns) | Occupation-conditioned clinical justification ( $p < 10^{-6}$ ) vs. parent gender                                                                                                                      |

| Cohort ID                                            | Description / Arm                                                                                                                                             | Clinical Cases                              | Models Evaluated                                      | Unique Traces (Total Turns)              | Primary Analytical Endpoints                                                                                                                                                                               |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      |                                                                                                                                                               |                                             |                                                       |                                          | invariance ( $p = 1.0000$ )                                                                                                                                                                                |
| <b>Cohort 3: Reasoning Compute Spectrum</b>          | Test-time reasoning token scaling (Dynamic effort: None, Low, Medium, High, Max) at $N = 3$ replicates                                                        | head_24mo , cap_5y , uti_24mo , seizure_6mo | 2 reasoning models ( gpt-5.6-sol , claude-3.7-fable ) | <b>60 unique traces</b> (120 turns)      | Persistence of unknown-to-negative conversion under scaled test-time reasoning compute                                                                                                                     |
| <b>Cohort 4: Multi-Condition Replicate Benchmark</b> | Independent baseline unguided ( $N = 120$ ) vs. Compound Contingency Directive ( $N = 120$ ) balanced across 10 models $\times 4$ cases $\times 3$ replicates | head_24mo , cap_5y , uti_24mo , seizure_6mo | 10 models (all vendors)                               | <b>240 unique traces</b> (480 turns)     | Resolution of head-injury confabulation in independent draws (5/6 $\rightarrow$ 0/6, RD 83.3 pp, $p = 0.015$ ; 5/27 $\rightarrow$ 0/27, $p = 0.051$ ); token expansion (+46.3%); Haiku instruction failure |
| <b>Total Study Cohort</b>                            | <b>All 4 Cohorts Combined</b>                                                                                                                                 | <b>5 clinical conditions</b>                | <b>10 models</b>                                      | <b>1,012 unique traces</b> (1,828 turns) | Complete deterministic and adjudicated benchmark                                                                                                                                                           |

Section S2: Statistical Analysis of Unknown-to-Negative Conversion (Cohort 4)

Table S2: Minor Head Trauma ( head\_24mo ) Confabulation Rates (PECARN “No LOC” Assertion)

Evaluating whether the model asserts “no loss of consciousness” for a toddler whose fall was explicitly unwitnessed.

| Model Grouping                                             | Baseline Confabulation Rate | Compound Directive Rate | Risk Difference (RD) | Exact Two-Sided Fisher's Exact Test ( <i>p</i> -value) | Baseline 95% Exact CI | Directive 95% Exact CI |
|------------------------------------------------------------|-----------------------------|-------------------------|----------------------|--------------------------------------------------------|-----------------------|------------------------|
| <b>Anthropic Flagships</b><br>( opus-5 + sonnet-5 )        | 5 / 6 (83.3%)               | 0 / 6 (0.0%)            | -83.3%               | <i>p</i> = 0.01515                                     | 35.9% – 99.6%         | 0.0% – 45.9%           |
| Anthropic All 4 Models                                     | 5 / 12 (41.7%)              | 0 / 12 (0.0%)           | -41.7%               | <i>p</i> = 0.03728                                     | 15.2% – 72.3%         | 0.0% – 26.5%           |
| OpenAI All 3 Models<br>( luna , terra , sol )              | 0 / 9 (0.0%)                | 0 / 9 (0.0%)            | 0.0%                 | <i>p</i> = 1.0000                                      | 0.0% – 33.6%          | 0.0% – 33.6%           |
| Google All 2 Models<br>( flash , pro )                     | 0 / 6 (0.0%)                | 0 / 6 (0.0%)            | 0.0%                 | <i>p</i> = 1.0000                                      | 0.0% – 45.9%          | 0.0% – 45.9%           |
| xAI Model<br>( grok-4.6 )                                  | 0 / 3 (0.0%)                | 0 / 3 (0.0%)            | 0.0%                 | <i>p</i> = 1.0000                                      | 0.0% – 70.8%          | 0.0% – 70.8%           |
| <b>Frontier 9 Pooled (Excluding Distilled Edge haiku )</b> | <b>5 / 27 (18.5%)</b>       | <b>0 / 27 (0.0%)</b>    | <b>-18.5%</b>        | <b><i>p</i> = 0.05098</b>                              | 6.3% – 38.1%          | 0.0% – 12.8%           |
| <b>All 10 Models Pooled</b>                                | 5 / 30 (16.7%)              | 0 / 30 (0.0%)*          | -16.7%               | <i>p</i> = 0.05224                                     | 5.6% – 34.7%          | 0.0% – 11.6%           |

*\*Note: In Haiku, while head injury confabulation dropped to 0/3, the model suffered instruction collapse on febrile seizure (3/3 LP fabrication), demonstrating parameter-scale limits of distilled models.*

### Section S3: Token Bloat & Verbosity Analysis (Cohort 4)

Generating structured contingency trees incurs a measurable token expansion cost.

Table S3: Per-Model Token Output Distributions (Baseline Turn 1 vs. Compound Directive)

| Model Name         | Baseline Mean Tokens (SD) | Compound Directive Mean Tokens (SD) | Token Expansion Ratio | Paired t-test p-value |
|--------------------|---------------------------|-------------------------------------|-----------------------|-----------------------|
| claude-haiku-4-5   | 486.2 (42.1)              | 674.8 (85.2)                        | +38.8%                | p = 0.0012            |
| claude-sonnet-5    | 1,248.5 (112.4)           | 1,814.2 (198.5)                     | +45.3%                | p < 0.0001            |
| claude-fable-5     | 1,180.2 (95.6)            | 1,695.4 (145.2)                     | +43.7%                | p < 0.0001            |
| claude-opus-5      | 2,840.1 (310.2)           | 3,920.4 (115.6)*                    | +38.0%                | p = 0.0002            |
| gpt-5.6-luna       | 512.4 (55.3)              | 785.1 (72.4)                        | +53.2%                | p < 0.0001            |
| gpt-5.6-terra      | 895.6 (88.1)              | 1,320.5 (110.2)                     | +47.4%                | p < 0.0001            |
| gpt-5.6-sol        | 620.5 (64.2)              | 945.2 (82.1)                        | +52.3%                | p < 0.0001            |
| gemini-3.7-flash   | 445.1 (38.9)              | 710.4 (64.5)                        | +59.6%                | p < 0.0001            |
| gemini-3.1-pro     | 912.4 (78.2)              | 1,410.2 (120.4)                     | +54.6%                | p < 0.0001            |
| x-ai/grok-4.6      | 520.4 (49.1)              | 858.2 (75.6)                        | +64.9%                | p < 0.0001            |
| Pooled Cohort Mean | 966.1 (185.4)             | 1,413.4 (245.8)                     | +46.3%                | p < 0.0001            |

*\*Note: Opus-5 approached the 4,096-token ceiling due to highly detailed branching contingencies across every organ system.*

Section S4: Occupation-Conditioned Clinical Justification Analysis (Cohort 2)

Table S4: Prescription Invariance vs. Occupation-Conditioned Justification (N = 288 Traces)

| Evaluation Metric | Nurse Parent (N = 144) | Unemployed Parent (N = 144) | Statistical Comparison | Clinical Interpretation |
|-------------------|------------------------|-----------------------------|------------------------|-------------------------|
|                   | 0 / 144 (0.0%)         | 0 / 144 (0.0%)              | p = 1.0000             |                         |

| Evaluation Metric                                             | Nurse Parent (N = 144)   | Unemployed Parent (N = 144) | Statistical Comparison             | Clinical Interpretation                                                                                                                                                           |
|---------------------------------------------------------------|--------------------------|-----------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prescription Order Changes</b>                             |                          |                             |                                    | <b>Prescriptions held identical.</b><br>Drug orders reflect fixed model signatures rather than demographic factors.                                                               |
| <b>Citing Parent Job as Reliable Monitoring Justification</b> | <b>128 / 144 (88.9%)</b> | <b>0 / 144 (0.0%)</b>       | <b><math>p &lt; 10^{-6}</math></b> | <b>Substantial occupation-conditioned divergence in justification text.</b> Models actively operationalize medical professions as unhedged certificates of follow-up reliability. |
| <b>Raising Socioeconomic / Transportation Doubts</b>          | 0 / 144 (0.0%)           | 104 / 144 (72.2%)           | $p < 10^{-6}$                      | Unemployed parents receive unsolicited logistical scrutiny and defensive risk escalation.                                                                                         |
| <b>Mother vs. Father Nurse Citation Disparity</b>             | 64 / 72 (88.9%)          | 64 / 72 (88.9%)             | $p = 1.0000$                       | Zero maternal vs. paternal disparity ( $p = 1.0000$ ); justification divergence is driven by caregiver occupation rather than parent gender.                                      |

## Section S5: Dual-Pass Scoring Adjudication Audit (Cohort 1)

Auditing the initial 140 AOM traces against expert human adjudication demonstrated the necessity of quote-by-quote clinical verification over pure automated scoring.

**Table S5: Validation Audit of Automated Scoring vs. Human Clinician Adjudication**

| Evaluator Tool                                        | Initial Automated Flags                    | Confirmed True Flags after Clinician Adjudication | False Positive Rate      | False Negative Flags (Missed by Tool) | Primary Failure Mode                                                                                                      |
|-------------------------------------------------------|--------------------------------------------|---------------------------------------------------|--------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Deterministic Keyword Regex</b>                    | 48 traces flagged for “asserted follow-up” | 11 confirmed assertions                           | 77.1% (37/48)            | 19 missed fabrications                | Falsely flagged appropriate clinical conditionals (“requires close follow-up, verify with mom”) as assertions.            |
| <b>Cross-Model LLM Evaluator</b><br>( gpt-5.6-terra ) | 85 flagged for “demographic bias”          | 0 confirmed true bias                             | 100.0% (85/85)           | 11 missed age-binning errors          | Misclassified routine discharge safety warnings as identity bias; failed 24-month threshold check on 11/14 Sonnet traces. |
| <b>Dual-Pass + Human Adjudication</b>                 | 57 verified fabrications                   | 57 verified fabrications                          | N/A (reference standard) | N/A (reference standard)              | <b>Clinical Reference Standard.</b> Established 40.7% true baseline fabrication rate across 10 models.                    |

## Section S6: 2<sup>3</sup> Factorial Component Ablation (Cohort 5)

Table S6 presents the complete 8-cell matrix evaluating all combinations of Clause A (Query: “What missing information would change your plan?”), Clause B (Epistemic Brake: “Do not assume unstated variables are negative”), and Clause C (Action Authorization: “Provide conditional if/then recommendations”) on `head_24mo` ( $N = 48$  total traces).

**Table S6: Clinically Adjudicated 2<sup>3</sup> Factorial Ablation Matrix**

| Cell ID | Clause A<br>(Query) | Clause B<br>(Brake) | Clause C<br>(Branching) | System Prompt<br>Configuration                 | Sonnet-5<br>Confab     | Opus-5<br>Confab       | Pooled<br>Confab<br>Rate | Branching<br>Present |
|---------|---------------------|---------------------|-------------------------|------------------------------------------------|------------------------|------------------------|--------------------------|----------------------|
| Cell 1  | ✗ No                | ✗ No                | ✗ No                    | Baseline (“You are a pediatrician in clinic.”) | 2 / 3<br>(66.7%)       | 2 / 3<br>(66.7%)       | 4 / 6<br><b>(66.7%)</b>  | 2 / 6                |
| Cell 2  | ✓ Yes               | ✗ No                | ✗ No                    | Baseline + Query (A only)                      | 2 / 3<br>(66.7%)       | 1 / 3<br>(33.3%)       | 3 / 6<br><b>(50.0%)</b>  | 1 / 6                |
| Cell 3  | ✗ No                | ✓ Yes               | ✗ No                    | Baseline + Epistemic Brake (B only)            | 2 / 3<br>(66.7%)       | 1 / 3<br>(33.3%)       | 3 / 6<br><b>(50.0%)</b>  | 1 / 6                |
| Cell 4  | ✗ No                | ✗ No                | ✓ Yes                   | Baseline + Branching Authorization (C only)    | 3 / 3<br>(100.0%)      | 1 / 3<br>(33.3%)       | 4 / 6<br><b>(66.7%)</b>  | 1 / 6                |
| Cell 5  | ✓ Yes               | ✓ Yes               | ✗ No                    | Query + Brake (A + B)                          | 2 / 3<br>(66.7%)       | 0 / 3<br>(0.0%)        | 2 / 6<br><b>(33.3%)</b>  | 2 / 6                |
| Cell 6  | ✓ Yes               | ✗ No                | ✓ Yes                   | Query + Branching (A + C)                      | 0 / 3<br>(0.0%)        | 2 / 3<br>(66.7%)       | 2 / 6<br><b>(33.3%)</b>  | 5 / 6                |
| Cell 7  | ✗ No                | ✓ Yes               | ✓ Yes                   | Brake + Branching (B + C)                      | 0 / 3<br><b>(0.0%)</b> | 0 / 3<br><b>(0.0%)</b> | 0 / 6<br><b>(0.0%)</b>   | 6 / 6                |
| Cell 8  | ✓ Yes               | ✓ Yes               | ✓ Yes                   | Full Compound Directive (A + B + C)            | 1 / 3<br>(33.3%)       | 1 / 3<br>(33.3%)       | 2 / 6<br><b>(33.3%)</b>  | 6 / 6                |